

Ma Huamin

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Computer Science and Technology

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EDUCATION

- **Undergraduate in Computer Science and Technology** (2024 – Present)
Sichuan University, Chengdu, China
GPA: 3.71/4.0 (Top 10%)
- **Core Courses**
Artificial Intelligence Foundation(94), Probability & Statistics (93), Calculus (95), Data Structures & Algorithms (92), Linear Algebra (91), Discrete Mathematics (88), OOP(C++) (88)
- **English Proficiency**
CET-4: 522 CET-6: 456

HONORS & AWARDS

- National Endeavor Scholarship (2025)
- University First-Class Individual Scholarship (2025)

PROJECTS

- **Stage-wise NWP Integration for Wind Power Forecasting** 2026
Principal Investigator | Undergraduate Innovation Training Program
Investigated the role of Numerical Weather Prediction (NWP) information at different stages of wind power forecasting, addressing the lack of a unified understanding of NWP integration strategies. Proposed a stage-wise framework that incorporates NWP through representation learning, latent-level fusion, and forecast correction, together with an NWP bias-correction strategy to improve forecast quality. Conducted extensive experiments across multiple forecasting horizons, NWP error levels, and wind fluctuation scenarios to evaluate the strengths and limitations of different integration stages. Preliminary results demonstrate that the choice of NWP integration stage significantly affects forecasting accuracy and model robustness, providing practical guidance for the design of NWP-aware wind power forecasting models.
- **NIPT-Based Fetal Anomaly Detection & Optimal Screening Timing** 2025
Programmer | China Undergraduate Mathematical Contest in Modeling (CUMCM)
To address a noninvasive prenatal testing optimization problem in the 2025 CUMCM, 1) *multiple regression* was used to model and test the relationship between DNA concentration, gestational age, and BMI; 2) *K-means clustering* was employed to optimize the timing through BMI grouping; 3) *random survival forest* was utilized to integrate age and height for predicting success rates, optimizing the detection window and handling censored data; 4) *random forest classification* was applied to achieve high-dimensional discrimination of trisomy in female fetuses.

SELF-ASSESSMENT

- **Responsible and reliable.** Accurately and efficiently complete tasks assigned by supervisors or senior fellow students; proactively consult relevant literature and strive to resolve issues independently.

- **Self-motivated.** Regularly follow cutting-edge developments and recent papers, keeping an eye on open problems in the field. Actively explore new ideas and techniques.
- **Strong logical thinking and solid mathematical foundation.** Possess an in-depth understanding of the core mathematical principles behind common machine learning models, and excel at systematically organizing data flows and algorithmic pipelines.
- **Meticulous and patient.** Capable of independently setting up experimental environments, tuning hyperparameters, and interpreting results; persistent in refining models through repeated debugging.
- **Proven research experience.** Served as principal investigator of an undergraduate innovation project, leading comparative experiments and contributing to paper writing, which honed both execution and teamwork skills.